

BOWLING GREEN 21 NNE, KY, USA

WMO: 724130

Lat: 37.2504N

Lon: 86.2325W

Elev: 790

StdP: 14.28

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63849

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.5	15.3	-0.5	5.5	15.6	4.3	7.1	17.6	9.6	54.4	8.7	53.6	1.0	N/A	0.332

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	91.1	74.9	88.9	74.6	87.0	73.5	77.7	86.8	76.6	85.4	75.4	83.9	3.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.1	135.6	83.2	73.8	129.6	81.6	72.8	125.3	80.5	41.7	86.8	40.5	85.7	39.4	83.9	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	6.9	6.0	4.8	94.6	5.4	4.2	0.9	97.7	-2.2	100.1	-5.2	102.5	-9.1	105.6
			3.6	79.6	5.8	1.5	-0.5	80.7	-3.9	81.6	-7.2	82.4	-11.4	83.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.9	35.6	39.0	48.6	59.0	66.9	74.1	76.4	76.1	70.6	59.4	48.0	40.1
	DBStd	17.08	12.25	12.08	11.31	9.52	7.83	4.73	4.33	4.71	6.58	9.27	9.73	10.46
	HDD50	1524	469	343	168	30	1	0	0	0	0	22	150	341
	HDD65	4001	912	728	519	221	79	3	1	1	26	224	514	773
	CDD50	4407	22	36	124	301	524	723	817	808	617	313	89	33
	CDD65	1411	0	1	10	42	137	277	353	345	193	50	3	0
	CDH74	10999	0	2	60	301	967	2159	2923	2807	1424	343	13	0
	CDH80	3627	0	0	6	40	227	693	1063	1049	470	78	1	0
Wind	WSAvg	2.3	2.8	2.8	3.1	3.2	2.5	2.0	1.7	1.6	1.6	1.9	2.3	2.5
Precipitation	PrecAvg	50.7	4.1	3.7	5.0	4.4	5.6	4.3	4.4	3.3	3.5	3.8	3.6	4.8
	PrecMax	62.0	8.0	6.8	14.4	7.9	12.2	8.0	8.1	9.1	8.0	8.5	6.5	10.6
	PrecMin	39.1	2.2	1.3	2.4	1.9	2.5	2.1	1.4	0.6	1.0	0.4	0.9	1.9
	PrecStd	5.8	1.3	1.6	2.8	1.7	2.6	1.7	1.6	1.7	2.2	2.2	1.3	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.2	71.8	80.2	84.1	87.5	91.0	93.9	95.3	92.7	86.5	75.3	68.5
		MCWB	58.8	59.8	62.4	64.8	71.6	74.2	76.0	78.1	72.2	71.3	60.1	60.8
	2%	DB	62.1	66.4	75.0	80.6	84.9	88.6	90.2	91.5	88.4	81.8	70.9	64.2
		MCWB	55.9	57.2	59.6	62.7	69.3	74.0	76.2	75.7	72.3	66.5	58.7	57.5
	5%	DB	59.3	62.3	70.6	77.7	82.8	86.8	88.3	88.4	85.7	77.9	66.7	60.2
		MCWB	53.5	55.0	57.1	61.1	68.4	73.3	75.7	74.1	71.4	64.3	56.5	55.0
	10%	DB	55.1	57.9	66.2	74.3	80.1	84.3	86.1	86.1	82.2	73.6	62.8	56.4
		MCWB	48.7	51.6	54.0	60.2	67.8	72.3	74.8	73.0	70.1	63.7	53.6	51.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.4	62.6	64.1	68.6	74.2	77.9	80.0	79.0	76.2	73.8	64.3	63.2
		MCDB	64.9	68.9	75.8	79.8	83.6	86.2	88.7	89.4	85.3	82.3	71.0	67.2
	2%	WB	56.8	60.3	61.7	66.7	72.4	76.4	78.2	77.0	74.6	70.8	61.7	59.6
		MCDB	60.4	66.1	72.1	76.9	81.9	85.7	87.2	85.5	84.2	77.1	67.1	64.0
	5%	WB	53.0	57.3	59.4	64.5	70.9	75.2	77.1	75.7	73.1	67.8	58.5	56.4
		MCDB	56.8	61.0	67.2	73.2	80.1	84.3	86.2	84.0	82.2	74.3	63.9	60.0
	10%	WB	48.8	53.2	56.7	62.3	69.4	73.8	75.9	74.5	71.6	64.3	54.9	52.4
		MCDB	53.0	57.9	64.0	70.8	78.1	82.4	84.3	82.2	79.4	71.8	60.8	56.5
Mean Daily Temperature Range	5% DB	MDBR	15.5	16.7	18.5	19.9	18.2	17.7	17.1	17.8	18.8	19.2	17.7	14.7
		MCDBR	18.3	19.7	22.2	21.8	20.1	19.6	19.1	20.3	21.3	21.7	20.8	17.7
		MCWBR	15.5	14.7	13.3	10.6	9.3	9.0	8.3	8.6	8.5	10.0	11.8	13.0
	5% WB	MCDBR	18.6	18.3	19.6	18.9	17.6	18.0	17.3	16.8	18.0	18.0	17.0	16.5
		MCWBR	16.4	15.5	12.0	11.1	9.0	9.2	8.5	8.1	8.1	10.3	12.8	14.3
Clear-Sky Solar Irradiance	taub	0.312	0.321	0.361	0.405	0.443	0.466	0.497	0.510	0.422	0.362	0.330	0.306	
	taud	2.470	2.425	2.369	2.244	2.183	2.149	2.081	2.029	2.260	2.421	2.467	2.493	
	Ebn at Noon	278	289	286	278	268	261	252	245	262	271	269	272	
	Edh at Noon	26	31	36	43	47	48	51	53	40	31	26	24	
All-Sky Solar Radiation	RadAvg	647	849	1195	1585	1770	1982	1886	1751	1461	1110	777	558	
	RadStd	67	114	114	113	161	114	111	96	129	126	82	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+227	+148

Nomenclature: See separate page